

Homework I

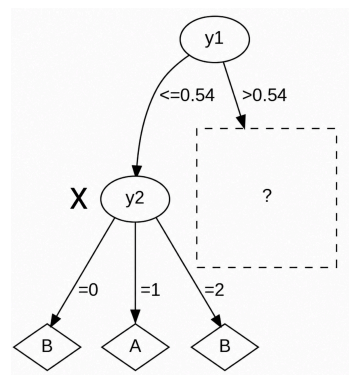
Deadline: via Fenix as PDF

- Submit Gxxx.ZIP in Fenix where xxx is your group number. The ZIP should contain two files: Gxxx_report.pdf with your report and Gxxx_notebook.ipynb with your notebook demo according to the suggested templates
- It is possible to submit several times on Fenix to prevent last-minute problems. Yet, only the last submission is kept
- Exchange of ideas is encouraged. Yet, if copy is detected after automatic or manual clearance, homework is nullified and IST guidelines apply for content sharers and consumers, irrespectively of the underlying intent
- Please consult the FAQ before posting questions to your faculty hosts

I. Pen-and-paper [9v]

Consider the partially learnt decision tree from the dataset D . D is described by 4 input variables (1 numeric and 3 categorical) and a target variable with three classes.

D	y_1	y_2	y_3	y_4	y_{out}
x_1	0.10	0	0	0	B
x_2	0.22	1	0	0	A
x_3	0.30	0	1	0	B
x_4	0.40	2	0	0	A
x_5	0.44	2	0	1	B
x_6	0.48	2	1	2	B
x_7	0.60	0	0	0	C
x_8	0.68	0	0	1	C
x_9	0.75	0	0	0	A
x_{10}	0.82	0	1	2	B
x_{11}	0.88	0	1	2	A
x_{12}	0.95	1	2	2	C



- 1) [5v] Complete the given partially learnt decision tree using information gain (Shannon entropy with $\log_2$), considering that i) a minimum of 3 observations is required to split an internal node, ii) decisions by ascending alphabetic order should be placed in case of ties, and iii) y_1 , already used in the first split, should not be considered for subsequent splits.
- 2) [2v] Draw the training confusion matrix for the learnt decision tree.
- 3) [2v] Perform post-pruning on the left branch of the decision tree by determining whether the subtree rooted at node X (indicated in the tree) can be replaced by a leaf

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node while allowing a decrease in accuracy of up to 10 percentage points, measured on the entire training set.

II. Programming [11v]

Consider the `breast_cancer.csv` dataset, available at the homework tab, comprising 30 biological features used to classify 569 patients into two classes: malignant (coded as M) and benign (coded as B).

1. [1.5v] Using `f_classif` from `sklearn`, compute the F-statistic and p-value for each input variable. Rank the variables according to their discriminative power (from most to least discriminative) and remove variables whose p-value is above 0.05.
2. [3.5v] Using a stratified 80-20 training-testing split, train decision trees with minimum sample split in $\{2, 5, 10, 20, 50, 100\}$ and assess their training and testing accuracies in a single plot. Use `random_state=1` for both the training-testing split and the decision tree and keep the remaining parameters at their default values. Identify and plot the best-performing tree. In case of a tie between multiple settings, select the tree with the larger `min_samples_split`.
3. [2v] Critically analyze these results, including the model's generalization capacity across the different minimum sample split settings.
4. [1.5v] A clinic asks you to develop a model for the initial screening of breast cancer, which will be used to help identify patients who should be referred for further clinical assessment.

Suppose two candidate models achieve the following results on the test set:

- Model A: Precision = 98%, Recall = 88%
- Model B: Precision = 88%, Recall = 98%

Which model would you recommend to the clinic? Justify your choice based on the clinical context.

5. [2.5v] Apply the TreeSHAP approach using the `shap.TreeExplainer` class to compute the SHAP values. Apply this to the test set using the best tree-based model obtained in Question 2 and generate the required plots, following the documentation provided below.
 - a. Identify the two features with the greatest overall impact on the model's predicted probability of malignant breast cancer using `shap.plots.bar`.

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- b. Using `shap.plots.beeswarm`, analyse the two features identified in a). For each feature, determine whether high or low feature values tend to increase or decrease the model's predicted probability of malignant breast cancer, and interpret the results in the context of the problem.

Documentation:

<https://shap.readthedocs.io/en/latest/generated/shap.TreeExplainer.html>

END